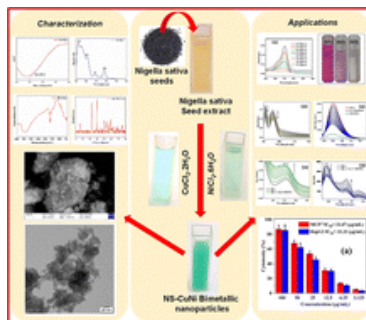


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From the journal:

New Journal of Chemistry**Green synthesis of bimetallic Cu–Ni nanoparticles using *Nigella sativa* seed extracts: morphological characterization, anticancer potential, biomolecular binding, and photocatalytic dye degradation**[Subramani Santhiya](#),^a [Paulraj Adwin Jose](#),^b [Murugesan Sankarganesh](#), ^{*a} [Ganesan Jeevan](#),^c [Shunmugasundaram Gurusamy](#),^d and [Senthil Kumar Raju](#)^e [Author affiliations](#)**Abstract**

This study reports the green synthesis, comprehensive characterization, DNA and BSA binding assessment, and photocatalytic dye degradation activity of bimetallic copper–nickel nanoparticles (NS–CuNi) synthesized using water-soluble *Nigella sativa* seed extracts. The eco-friendly, plant-mediated approach utilizes phytochemicals as both reducing and stabilizing agents, eliminating the need for toxic chemicals and enhancing the nanoparticles' biological compatibility. NS–CuNi nanoparticles with high elemental purity and surface functionalities were derived from the seed extract. Optical studies revealed distinct dual surface plasmon resonance peaks (260 and 325 nm), while XRD and electron microscopy evidenced a face-centered cubic morphology, narrow size distributions, and high surface area. The nanoparticles exhibited remarkable photocatalytic performance, achieving over 94% degradation of hazardous rhodamine B and methylene blue dyes within 70–80 minutes under visible light, with pseudo–first-order kinetic behavior (rate constants: 0.0405 min^{−1} for rhodamine B and 0.038 min^{−1} for methylene blue). Biomolecular interaction studies demonstrated strong DNA binding affinity (5.85 × 10⁵ M^{−1}) and reversible, moderate BSA interaction (K_{sv} : 8.1 × 10⁴ M^{−1}), supporting their biomedical compatibility. *In vitro* cytotoxicity assays revealed significant dose-dependent inhibition of MCF7 and HepG2 cancer cells, with IC₅₀ values (23.47 and 32.32 μg mL^{−1}, respectively) comparable to other green-synthesized bimetallic systems, though cisplatin exhibited superior potency. The synergistic effects of copper and nickel, combined with the natural capping biomolecules, imparted enhanced redox activity, bio-interactivity, and catalytic efficiency.



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